

Compressed Traffic Assignment: Numerical Report v4.1

Certified results of the lockstep campaign (2026-07-25 – 2026-07-27)

Campaign record — `campaign.weighted/`, branch `version.3`

27 July 2026

Headline, Chicago Sketch (path-rich regime). On the nested-pool ladder with a fixed set of 19,112 OD pairs (94.2% of demand) and matched ALM operators, compression with a rank-6 backbone and at most 500 dynamically repriced OD exchange atoms attains

$$S(G^*=0.015) = \mathbf{1.87\times} \quad (\bar{K} = 31.6, n = 603,232), \quad S(G^*=0.015) = 1.46 \times \quad (\bar{K} = 25.4),$$

at *matched time-to-accuracy against a converging full-space baseline*. The advantage rises monotonically with route richness and with target tightness inside the reachable band; below $\bar{K} \approx 15$ compression does not pay ($S = 0.81$ at $\bar{K} = 14.6$; does not reach the targets at $\bar{K} = 7.7$).

Headline, Chicago Regional (structural negative, explained). The E0 pool has $\bar{K} = 7.48$, so the reduction ceiling (11) caps compression at $1 - 1/7.48 = 86.6\%$ — exactly the measured saturation value ($84.0 \rightarrow 86.3 \rightarrow 86.6 \rightarrow 86.6\%$ for $\tau = 1.03, 3, 8, 20$; at $\tau=20$ the split retains 270,716 majors against 270,598 OD pairs, i.e. one path per OD, the hard floor). The compiled solver paid only above $\sim 90\%$ reduction, hence Regional *cannot* pay with this pool for any (τ, r) : the certified C++ bound is $S < 0.58\times$. The negative is a property of the pool’s richness, not of the method or its tuning, and is predictable in advance from two integers (n and ℓ).

[Note to co-authors: Everything in this report traces to CSVs in `campaign.weighted/results/` (commits `f6deb4e`, `4612cae`, `a25b5a4`, `a01413d`, `165e816`). All timings single-process, single-thread. The old “ $3.26\times$ with essentially zero accuracy loss” phrasing is withdrawn; see §7.]

1 Model, algorithms, and metrics

Assignment model. Path flows $x \in \mathbb{R}_{\geq 0}^n$ over a pool of n paths and ℓ OD pairs, incidence $B \in \{0, 1\}^{n \times m}$, OD aggregation $A \in \{0, 1\}^{\ell \times n}$, demand $d \in \mathbb{R}^\ell$, BPR link costs $t(v)$, $v = v_0 + B^\top x$. Beckmann objective

$$Z(x) = \sum_{a=1}^m \int_0^{v_a} t_a(s) ds. \tag{1}$$

Operator (matched on both sides). SPG-ALM: at outer iteration k , minimise over the representation’s variables

$$\mathcal{L}_\rho(x, \lambda) = Z(x) + \lambda^\top (Ax - d) + \frac{\rho}{2} \|Ax - d\|_2^2, \quad \lambda \leftarrow \lambda + \rho (Ax - d), \tag{2}$$

inner solver L-BFGS-B (maxiter 200, gtol 10^{-5}), $\rho_0 = 100$ doubled when the constraint residual fails to contract by $4\times$, identical schedules in every arm.

Compressed representation (incumbent). Major/minor split at threshold τ (every OD keeps its maximum-flow path explicit); minors are decoded through a global signed basis with an affine anchor:

$$x_{\mathcal{M}} = w_0 + U_r z, \quad \text{feasibility } w_0 + U_r z \geq 0 \text{ (ALM hinge, multiplier } \mu), \quad (3)$$

U_r = leading left singular vectors of SB_2 , $S = \text{diag}(\sqrt{w_0} + \varepsilon)$ (flow-weighted *row-scaling heuristic*; it is *not* the minimiser of the flow-weighted reconstruction loss, which is $\text{span}(S^{-1}\tilde{U}_r)$ — verified 2026-07-25 and stated wherever the basis is described).

Hybrid global–local representation (new). A low-rank backbone plus sparse, OD-local, *coefficient-bounded* route-exchange atoms on an active set $\mathcal{A}$:

$$x_{\mathcal{M}} = w_0 + U_G z_G + \sum_{q \in \mathcal{A}} \alpha_q a_q, \quad a_q = \frac{1}{\sqrt{2}}(e_{p_q^{\text{low}}} - e_{p_q^{\text{high}}}), \quad \mathbf{1}^\top a_q = 0, \quad (4)$$

where $p_q^{\text{low}}/p_q^{\text{high}}$ are the cheapest/costliest *used* minors ($x > \varepsilon$) of OD q at current costs, and

$$-\sqrt{2}w_0[p_q^{\text{low}}] \leq \alpha_q \leq \sqrt{2}w_0[p_q^{\text{high}}] \quad (5)$$

so the atom part alone cannot violate nonnegativity: atoms encode *flow exchange*, not signed reconstruction. The active set is chosen by per-OD disequilibrium

$$G_q = \sum_{p \in P_q} x_p (c_p - c_q^{\min}), \quad \mathcal{A} = \{q : G_q \text{ among the top } B\}, \quad (6)$$

and, in the *dynamic* variant, the anchor w_0 , the costs, G_q and the atom pairs are re-computed every outer iteration (equilibrium-error-guided column generation in the tangent space); λ, ρ carry across repricings.

Independent evaluation (the third evaluator). Every claim uses one independently implemented full-space scorer, applied to the raw decoded flows of *both* solvers — never a solver’s internal gap:

$$G(x) = \frac{\sum_q \sum_{p \in P_q} x_p (c_p - c_q^{\min})}{\sum_q d_q c_q^{\min}} \quad (\text{common equilibrium gap, pool-min costs}), \quad (7)$$

$$r_{\text{OD}} = \frac{\sum_q |\sum_{p \in P_q} x_p - d_q|}{\sum_q d_q}, \quad r_{\text{OD}, \max} = \max_q \frac{|\sum_p x_p - d_q|}{\max(d_q, \epsilon)}, \quad (8)$$

$$r_{\text{PG}} = \frac{\|x - \Pi_{\mathcal{X}}(x - \alpha \nabla Z(x))\|_2}{\max(1, \|x\|_2)}, \quad \alpha = 10^{-2} \quad (\Pi_{\mathcal{X}}: \text{per-OD Euclidean simplex proj.}). \quad (9)$$

Speed claims are *time-to-target*

$$S(G^*) = \frac{T_{\text{full}}(G^*)}{T_{\text{comp}}(G^*)}, \quad (10)$$

never final wall time after a solver has stalled. Repeated-scenario economics use $H^* = \lceil T_{\text{prep}} / (T_{\text{warm}} - T_{\text{comp}}) \rceil$.

2 Result 1: the reduction ceiling

Every OD pair must keep at least one explicit path, so route richness $\bar{K} = n/\ell$ caps the dimensional reduction *before any tuning*:

$$\text{reduction} \leq \frac{n - \ell - r}{n} \approx 1 - \frac{1}{\bar{K}}. \quad (11)$$

Table 1: The ceiling binds everywhere (exp06; `exp06_regional_tau.csv`).

instance	$\bar{K}$	ceiling $1-1/\bar{K}$	measured reduction	note
Regional-E0 ($\tau=1.03$)	7.48	86.6%	84.0%	
Regional-E0 ($\tau=3$)	7.48	86.6%	86.3%	
Regional-E0 ($\tau=8, 20$)	7.48	86.6%	86.6%	saturated: 270,716 majors vs 270,598 ODs
Sketch-K10	11.09	91.0%	89.8%	
Sketch-K15	15.34	93.5%	92.4%	

The compiled solver paid only above $\sim 90\%$ reduction; Regional-E0 cannot reach it, hence the certified bound $S < 0.58 \times$ (full C++ solve 1085.3s, objective 18,854,545.32 reproduced bit-for-bit; compressed stopped at >1860 s). The old Regional negative is thereby *explained and predictable*: a network needs $\bar{K} \gtrsim 10\text{--}15$ before compiled compression can be expected to pay.

3 Result 2: the representation floor

Instrument. Full and compressed ALM advance *one outer iteration at a time* from a byte-identical iteration 0 ($w_0 = x_{\mathcal{M}}^0$, $z^0 = 0$, $y^0 = x_{\text{maj}}^0$; identity gate: all differences exactly 0.0), scored each step by (7)–(10) (exp08, Chicago Sketch K10, scenario d05 = demand $\times (1 + U(-0.05, 0.05))$, seed 100, $r = 6$).

Table 2: Ten synchronized outer iterations (abridged; `exp08_lockstep_K10_d05_r6.csv`).

k	G_{full}	G_{comp}	$r_{\text{PG,full}}$	$r_{\text{PG,comp}}$	cpu _{full} (s)	cpu _{comp} (s)
1	0.00503	0.01563	0.00278	0.00275	41.5	15.6
4	0.00110	0.01358	0.00032	0.00105	37.5	14.2
7	0.00074	0.01370	0.00009	0.00103	39.6	21.3
10	0.00069	0.01338	0.00003	0.00102	50.8	1.3

OD feasibility is comparable throughout; the divergence is purely equilibrium quality. Kernel accounting: the full side spends 96% of CPU in gradient + link aggregation; the compressed side spends **74%** in the dense $O(n_{\mathcal{M}}r)$ decode + hinge; the OD loop is 3–4% on both sides. A $10\times$ variable reduction delivers only $2.4\times$ per-outer cost.

Five controls (exp09), branched from the same $k=4$ state.

- C1 *Shared duals* (both directions): full escapes under either side’s λ ($G = 0.00091/0.00092$); compressed stays at its plateau (0.0132/0.0138). Multiplier drift eliminated.
- C2 *Hinge accounting*: at the plateau the iterate sits on the box boundary ($\min u = -1.1 \times 10^{-2}$, $H = 4.7 \times 10^3$, $\|\nabla_z H\| = 2.3 \times 10^3$); removing the hinge yields $G = 2.22$. The hinge is *necessary but not the root cause*: the feasible set is $\text{span}(U) \cap \{w_0 + Uz \geq 0\}$.

- C3 *Gradient geometry*: with g_T = per-OD-centred path costs, $\eta_{\text{captured}} = 0.134$, $\eta_{\text{missing}} = 0.866$; 87.7% of $\|g_T\|^2$ lies in the minor block, of which $\text{span}(U_6)$ captures **1.26%** (measured at the $r=6$ plateau; not extrapolated to other ranks).
- C4 *Full-space escape* from the identical iterate and ALM state: G 0.0135 $\rightarrow$ 0.00277 in *one* step, while a paired compressed step does nothing (0.0138). Decisive.
- C5 *Rank replay*: plateaus 0.0134/0.0135/0.0127 for $r = 6/10/20$ — the limitation persists over the tested rank range.

$$G_{\min}(U, r) \approx 1.3 \times 10^{-2} \text{ for the tested global-SVD representation, } r \in \{6, 10, 20\} \quad (12)$$

Replication: on K15-d05 the plateau is 0.0162, the escape step gives $0.01618 \rightarrow 0.00415$, and $\eta(\text{minor}) = 1.58\%$ — same mechanism, second pool. *Mechanism (exact wording)*: the global latent basis poorly aligns with the remaining OD-specific equilibrium directions, and the small aligned component is further restricted by the nonnegativity boundary of the decoded minor flows.

4 Result 3: structured representations and dynamic atoms

Table 3: One ALM step from the identical K10-d05 plateau point, same (λ, ρ) (exp10); then full trajectories from iteration 0 (exp10b/exp11). $G_0 = 0.01366$.

representation	R	η_{minor}	G after 1 step	cpu (s)	trajectory plateau
global $r=6$ (incumbent)	6	1.27%	0.01094	28.5	0.01352
origin blocks, light	1,081	4.10%	0.00819	28.5	0.01132
origin blocks, medium	2,792	9.07%	0.00776	34.5	0.01123
odx1 (flow-ranked atoms)	69,707	5.06%	0.01167	33.8	—
odx1g (cost-selected, static)	69,707	—	—	—	0.01169 (converges cleanly)
odxfull (within-OD zero-sum)	731,612	98.48%	0.00583	156.3	—
full-space escape (reference)	903,516	100%	0.00277	97.3	0.00069 ($k=10$)
static hybrid atoms, $B \leq 5000$	(exp11)	—	—	—	0.0103–0.0112
dynamic repricing , $B=2000$	$\sim 1,300$ active	—	—	—	0.00407, still descending

Three certified statements. (i) *OD-locality contains essentially all the missing gradient*: the within-OD zero-sum minor subspace captures 98.48% (major–minor exchange is only $\sim 1.5\%$): the floor is cross-OD coupling. (ii) *No cheap static locality removes the floor*: origin blocks plateau at ~ 0.0113 regardless of budget; statically priced atoms at any B plateau at 0.0103–0.0112. (iii) *Dynamic repricing is the mechanism*: rebuilding cost-selected, coefficient-bounded atoms (4)–(6) each outer reaches $G = 0.00407$ (still descending at the outer budget), passing 10^{-2} at 40s and 5×10^{-3} at 103s, with the active set self-pruning ($1,480 \rightarrow 1,158$) as ODs equilibrate. Stale atoms, not atoms, were weak. Exchange atoms also *converge cleanly* (inner iterations collapse to single digits; no boundary churn), unlike every signed-basis arm.

5 Catalogue: every representation tried, and why

Each candidate below was motivated by a specific failure of its predecessor; nothing was tried at random. All accuracy numbers are the common gap (7) on K10-d05 unless noted.

The through-line. Rows 1–4 established that the failure is not rank, weighting, or the anchor’s absence but the *class* (global coupling). Rows 5–9 localised it: origin granularity is too coarse at small budgets, OD granularity contains 98.5% of the missing gradient. Rows 7–8 fixed the *selection* rule (cost, not flow), row 10 fixed the *placement* rule (G_q -active ODs, coefficient bounds), and row 11 fixed the *timing* rule (reprice as costs move) — the only change that broke the floor. Row 12 marks the alternative route (change the feasibility geometry instead of the span), deliberately left to the Paper-2 line. [Note to co-authors: Rows 3 and 12 carry numbers from the pre-lockstep campaign (WO_LOAD_BEARING.md, highk_fixes.csv); engines stated there.]

6 Result 4: the break-even transition (Chicago Sketch ladder)

Fixed top-20,000 ODs by demand (94.2% of total volume; 19,112 after singleton removal), nested penalty-KSP pools $P_8 \subset P_{16} \subset P_{32} \subset P_{64}$ (dedupe; length filter $c_p \leq 2.5 c_q^{SP}$ — the topology supports $\bar{K}$ up to ~ 31), matched ALM pairs, pre-registered $G^* \in \{0.020, 0.015\}$ (exp12_extreme_M20000.csv). $S(G^*=0.015)$ is *monotone in richness* (DNS $\rightarrow$ DNS $\rightarrow 1.46 \rightarrow 1.87$): the advantage grows with the target inside the reachable band. Caveats, stated wherever these numbers appear: the τ split leaves 25% majors (75% reduction, not the $\sim 98\%$ extreme configuration — a tighter split is the obvious next lever and these S are likely understatements); Python engine; outer-granularity timing; one seed; controlled demand subset.

7 Requalified claims and retired claims

- **Say:** “On Chicago Sketch K15, the compiled compressed solver achieved $3.26\times$ (wall time) with a 0.046% Beckmann-objective difference *in the tested moderate-equilibrium-accuracy regime*; the lockstep evaluator subsequently showed both sides of that comparison operated at common gap $G \gtrsim 0.015$.” **Do not say:** “ $3.26\times$ with essentially zero accuracy loss.”
- **Say:** “Chicago Regional cannot pay with the E0 pool: $\bar{K} = 7.48$ caps reduction at 86.6% (11), below the $\sim 90\%$ paying regime; certified bound $S < 0.58\times$.” **Do not** present Regional as a tuning failure.
- **Say:** “Compression pays above a measurable richness threshold ($\bar{K} \approx 15\text{--}25$ here) at moderate accuracy targets, with $S(0.015)$ up to $1.87\times$, and provably cannot pay below the richness ceiling or beneath the representation floor (12).”
- Repeated scenarios (grid, Python): compressed 14.1s vs 50.1s warm-started full ($3.55\times$), $H^* = 2$; on K10 in C++ the warm-started full solver reaches its own floor almost instantly, so reuse claims are engine- and instance-specific.
- The flow-weighted basis is a row-scaling heuristic (kept because it improves the solved feasible gap in 27/34 paired comparisons), not reconstruction-optimal.
- Timing noise floor $1.45\times$: no smaller wall-time ratio is claimed anywhere.

Reproduction

```
# floor + controls          python drivers/exp08_lockstep.py --case K10 --scenario d05
                             python drivers/exp09_controls.py --case K10 --scenario d05
# structured representations python drivers/exp10_structured.py ; exp10b_trajectory.py
# hybrid + dynamic repricing python drivers/exp11_hybrid.py
```

```
# break-even ladder          python drivers/exp12_gen_pools.py --M 20000 --K 64 --ratio 2.5
                             python drivers/exp12_extreme.py --M 20000
```

Commits (branch `version_3`): `f6deb4e`, `4612cae`, `a25b5a4`, `a01413d`, `165e816`. Determinism caveat: ARPACK start vectors make U reproducible only up to column signs across processes; every controls driver therefore re-derives its branch state in-process.

Table 4: The representation ladder. “Capture” = η_{minor} at the $r=6$ plateau point; “floor” = trajectory plateau of G .

representation	construction	rationale	outcome / verdict
1. global SVD, un-weighted	U_r = top left sing. vectors of B_2	Eckart–Young optimal subspace for the incidence structure; the submitted method	works only above $G \approx 0.015$; retained as comparison arm
2. global SVD, flow-weighted	SVD of SB_2 , $S = \text{diag}(\sqrt{w_0} + \varepsilon)$	high-flow minors should dominate the subspace	better solved gap in 27/34 pairs; NOT reconstruction-optimal (that is $\text{span}(S^{-1}\tilde{U}_r)$); campaign default
3. offset-free ($w_0=0$) ablation	exact removal: $v \leftarrow v - B_2^\top w_0$, $d \leftarrow d + A_2 w_0$	is the anchor a convenience or a mechanism?	anchor is load-bearing : grid mean speedup $2.60\times \rightarrow 1.07\times$, Sioux $6.2\times \rightarrow 1.34\times$
4. rank ladder $r=6/10/20/50$	same class, more modes	maybe the floor is just rank starvation	floor 0.0134/0.0135/0.0127: rank-insensitive ; rank is a cost dial ($2r \lesssim \text{links/path}$), not an accuracy dial
5. origin-block SVD (light/med)	$U = \text{blkdiag}(U_1, \dots, U_{ O })$, $r_o = \lceil \alpha \sqrt{N_o} \rceil$, $R=1081/2792$	C3 showed cross-OD coupling: block by origin so corrections cannot disturb unrelated origins; SP trees/generation are origin-organised	capture 4.1/9.1%; dominates global on the one-step frontier at equal CPU, but class-level floor ~ 0.0113 at both budgets
6. destination-block SVD	(designed, deliberately not run)	useful only for destination-concentrated demand; origin ordering preferred for SP trees, per-origin pricing, parallelisation	deferred as a comparison arm
7. odx1: flow-ranked exchange atoms	one $a_q = (e_j - e_{j'})/\sqrt{2}$ per OD, two <i>largest-flow</i> minors	$\mathbf{1}^\top a_q = 0$: conservation automatic; substitution not signed reconstruction	capture only 5.06% with $R=69,707$: placement > count ; flow rank $\neq$ where the gradient is
8. odx1g: cost-selected static atoms	cheapest-vs-costliest <i>used</i> minor per OD at initial costs	steepest within-OD substitution direction	converges cleanly (no boundary churn, inner iters $\rightarrow$ single digits) to 0.0117: exchange class has only a capture defect — and static pairs go stale
9. odxfull: within-OD zero-sum basis	Helmert basis per OD, k_q-1 atoms, $R=731,612$	the OD-locality <i>upper bound</i> : what would full within-OD freedom buy?	capture 98.48% (major-minor exchange only $\sim 1.5\%$); proves the floor is cross-OD coupling; too large to be compression
10. static hybrid: backbone + top- B atoms	origin-light + atoms on top- B ODs by G_q (6), $B \in \{500, 2000, 5000\} \times 1, 2000 \times 2$; bounds (5)	concentrate scarce atoms where disequilibrium lives; box bounds remove hinge repair	plateaus 0.0103–0.0112; only $\sim 1/3$ of high-gap ODs even atom-eligible (≥ 2 used minors): staleness dominates
11. dynamic repricing	same as 10 but w_0 , costs, G_q , pairs rebuilt every outer; λ, ρ carry	equilibrium-error-guided column generation in the tangent space; the steepest pair <i>changes</i> as costs move	$G = 0.00407$ and descend-ing ; $t@10^{-2}=40$ s, $t@5 \times 10^{-3}=103$ s; active set self-prunes $1480 \rightarrow 1158$
12. grouped / box-bound (bundle)	nonnegative combination weights over grouped columns (earlier campaign, m3r1)	internalise the anchor: feasibility as a plain box, no signed crossing, no hinge	best objective of any operator on K10 (gap 0.0000) but $0.54\times$ in Python; never run in C++ — open Paper-2-adjacent arm

Table 5: Time-to-target and speedup (10). Compressed arm: global $r=6$ backbone + $B \leq 500$ dynamically repriced atoms (repriced on stall).

pool	$\bar{K}$	n	$T_{\text{full}}(.020)$	$T_{\text{comp}}(.020)$	$S(.020)$	$T_{\text{full}}(.015)$	$T_{\text{comp}}(.015)$	$S(.015)$
X8	7.7	147,088	11.0 s	DNS	—	20.9 s	DNS	—
X16	14.6	278,511	35.8 s	44.0 s	0.81	35.8 s	DNS	—
X32	25.4	484,580	59.6 s	44.7 s	1.33	86.3 s	59.3 s	1.46
X64	31.6	603,232	72.7 s	57.0 s	1.28	106.4 s	57.0 s	1.87